

Minguhn Kim

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Education

POSTECH (Pohang University of Science and Technology) Feb. 2025 – Feb. 2027 (Expected)
M.S. in Graduate School of Semiconductor Technology Pohang, South Korea

Laboratory: BEVIL (Broadband Electronics by Virtual Intelligence Lab)

Advisor: Prof. Byungsub Kim

- **Core Research:** AI-driven EDA: Automated Analog Layout Generation and RL-based Circuit Optimization.
- **Ongoing Work:** Developing CNN-based layout recognition modules to automate feature extraction and structural analysis of integrated circuits.

Chung-Ang University (CAU) Mar. 2019 – Feb. 2025
B.S. in Electrical and Electronic Engineering Seoul, South Korea

- **GPA: 4.34 / 4.50 (Summa Cum Laude)**
- **Military Service:** Completed duty in ROK Army 702nd Special Commando Battalion (Includes 2-year military leave of absence).

Key Research & Projects

Samsung AI Cluster Project 2025 – 2026
Primary Developer — DDI Layout Automation via A Search Algorithm*

- Engineered a GUI-based routing automation tool for Display Driver IC (DDI) layout designs, optimizing industrial workflows.
- **Leveraged AI-native development environments (Claude Code)** to accelerate complex software architecture design and algorithm prototyping.
- Implemented customized A* search logic to optimize metal routing paths, significantly reducing manual design overhead.
- Orchestrated software debugging and feature integration to enhance tool stability and design efficiency.

CTLE Optimization via Reinforcement Learning 2025
Research Contributor

- Supported the development of a Python-based framework to automate Continuous-Time Linear Equalizer (CTLE) parameter tuning.
- Assisted in applying Actor-Critic algorithms to bridge complex analog circuit behaviors with automated design flows.

Technical Skills

- **AI & Programming:** Python (PyTorch, NumPy, Pandas), Claude Code (AI-Assisted Development), C/C++ (Basic), Git.
- **Design Tools:** Cadence Virtuoso (Custom IC Design/Layout).
- **Algorithms:** Reinforcement Learning (Actor-Critic), CNN-based Layout Recognition, Graph Neural Networks, A* Search, Rule-based Automation.

Awards & Languages

- **Languages:** OPIc Advanced Low (AL) (Jan. 2026), NewTEPS 387 (Mar. 2024).
- **Academic Excellence Scholarship:** Chung-Ang University (Multiple-semester recipient).

Extracurricular Activities

- **Music Production & Songwriting:** Experience in composing and performing as a singer-songwriter using DAWs and piano; enjoys creative expression as a balance to technical research.
- **Physical Fitness:** Maintains a consistent routine of weight training and running for health and long-term stamina.